Sepehr Mahmoudian

SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI



EMAIL sepmhn@gmail.com

GITHUB github.com/sepahead LINKEDIN linkedin.com/in/sepehr-mahmoudian

Berlin-Brandenburg • available from Jan 2027 • German citizen • EN fluent / DE good / Persian native

PROFILE

Builds custom agent harnesses for domain-specific applications, knowledge graphs and Graph RAG; evaluates LLMs, VLMs, VLAs and world models with rigorous, evidence-first engineering.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction

CONTEXT ENGINEERING • DATA PROVENANCE • MODEL EVALUATION • REINFORCEMENT LEARNING

AI ENGINEERING

10+ YEARS

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

PROFESSIONAL EXPERIENCE

Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

Self-employed - PhD candidate at Technische Universität Darmstadt Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), Vision-Language-Action systems, VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

Senior AI Engineer

2026.01 - 2026.03

Neural Frames UG Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python pipelines for model evaluation and inference

AI Engineering Resident

2025.09 - 2026.01

Merantix AI Campus (Hacker Room) Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

Senior AI Engineer

2024.07 - 2025.07

Vision Labs UG Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

Research Associate

2021.07 - 2024.07

Freie Universität Berlin Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

Research Associate
2019.07 - 2020.06

Georg-August-Universität Göttingen | Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

Research Associate
2017.03 - 2019.07

Goethe-Universität Frankfurt | Frankfurt am Main, Germany · Simulation and statistical analysis of neural-network models

Visiting Scientist
2017.03 - 2019.03

Max Planck Institute for Dynamics and Self-Organization | Göttingen, Germany · Statistical methods for neural-network analysis

Research Associate
2015.02 - 2017.03

Forschungszentrum Jülich | Jülich, Germany · Neural-network simulations and reinforcement-learning experiments

Software Developer
2010.09 - 2012.03

Centre for Content Creation | Cyberjaya, Malaysia · Award-winning iPhone/iPad applications and digital products

EDUCATION

PhD Candidate (Dr. rer. nat.)
2020.04 - Present

Technische Universität Darmstadt | Darmstadt, Germany

- New mathematical PID analyses and information-theoretic representation analysis
- Engram's domain-specific AI agent produces provenance-tracked neural-model candidates for closed loop neural network simulations, with bounded compilation and simulation checks, in reproducible computational-neuroscience workflows

Master's by Research in Biosciences: Neuroscience
2012 - 2014

University College London | London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

Bachelor of Science (Honours) in Computer Science
2009 - 2012

Anglia Ruskin University | Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

SELECTED AWARDS & FUNDING

EDTH / 2025
Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub

HUMAN BRAIN PROJECT
Research funding

NEAR HORIZON
Confidential-computing grant

UCL & SHEFFIELD
Competitive academic scholarships

IOS APPLICATIONS
15 international product awards



SIGNATURE
Sepehr Mahmoudian

PLACE Berlin, Germany
DATE 10 October 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects
github.com/sepahead

HUGGING FACE Models & datasets
huggingface.co/torusprime

SELECTED PROJECTS

engram

★ 15 · RESEARCH PROTOTYPE

PROVENANCE-CONSTRAINED NEURAL R&D

Domain-specific AI agent turns qualified evidence into neural-model candidates, with bounded compilation and simulation checks.

Neural Networks · Knowledge Graphs · Closed-loop Sim

pid-rs

★ 15 · V0.9 SOURCE PRERELEASE

VERIFIABLE INFORMATION DECOMPOSITION

Measures how much information several signals share about an outcome, with built-in checks and provenance that keep results reproducible.

Rust / Python · Information Theory · Statistical ML

prisoma

★ 5 · V0.9 PRERELEASE / ACTIVE R&D

AUDITABLE CLOSED-LOOP EXPERIMENTS

Simulation environment for closed-loop VLA and world-model experiments, with capture, replay and provenance.

Rust / Python · VLA + World Models · Embodied Sim

NCP

★ 16 · V0.8 RELEASE / 1.0 RC BLOCKED

NEURAL CONTROL PROTOCOL

Links neural simulators, AI controllers and robots through four typed planes with local fail-safes and provenance.

Rust · Typed Control Planes · Zenoh

manwe

★ 6 · ALPHA

AIRSPACE PERCEPTION

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python · PyTorch · Multimodal Perception

crebain

★ 22 · RESEARCH PROTOTYPE

SPATIAL SITUATIONAL AWARENESS

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust · Sensor Fusion · Native Inference

melkor

★ 13 · HARDENING

GAUSSIAN-SPLAT INTEGRITY

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++ · Scene Reconstruction · Asset Validation

cobot-atlas
relief-atlas

★ 8 · ↓ 2,997 · PUBLISHED + DOI

AI-GENERATED SIMULATION DATASETS

2,023 robot-simulation and 10,079 disaster-relief GLB meshes with generation pipelines.

Python · Hugging Face · 3D / VLA

galadriel

★ 8 · RESEARCH PROTOTYPE

CROSS-SENSOR CONSISTENCY

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust · NIS / CUSUM · Signed Correlation

cortexel

★ 5 · 0.10.0-DEV.0

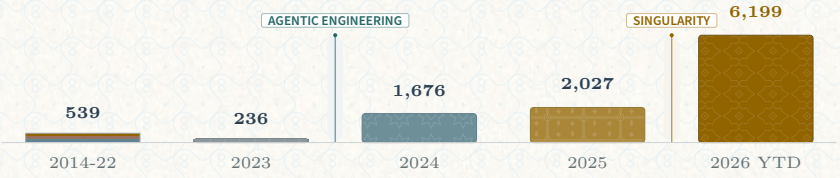
AUDITABLE NEURAL FIGURES

Checks scientific figures against declared contracts before rendering, so charts provably match their data and stay reproducible.

Figure Contracts · Deterministic SVG · Provenance

PORTFOLIO & ENGINEERING ACTIVITY

10,677	23	124	179
GITHUB CONTRIBUTIONS SINCE 2014	NON-FORK PUBLIC REPOSITORIES	STARS ON PUBLIC REPOSITORIES	ACTIVE DAYS IN 2026



CONTRIBUTION BY WEEKDAY



6,199 contributions • 2026 YTD

Mon 16.2% · Tue 13.9% · Wed 12.7% · Thu 9.7% · Fri 11.0% · Sat 19.0% · Sun 17.5%

CURRENT MOMENTUM

3.06× 2026 YTD contribution volume versus the full 2025 calendar year

6,199 YTD • 2,027 in 2025

TECHNICAL SKILLS

• AGENTIC AI	LLMs / VLMs | Agent Harnesses | AI Agents | Tool Use | Model Evaluation | Inference Pipelines
• MACHINE LEARNING	Python | PyTorch | Agentic Vision | Multimodal AI | Vision-Language-Action / VLA Evaluation
• KNOWLEDGE & RETRIEVAL	Knowledge Graphs | Graph RAG | RAG | Embeddings | Vector Search | Reranking | Data Provenance
• SYSTEMS & MLOPS	Rust | C++ | FastAPI | ONNX | TensorRT | CUDA | Metal / Core ML | Docker
• CLOUD & INFRASTRUCTURE	GCP | AWS | Azure | Kubernetes | Terraform | GitHub Actions | Linux | Zenoh